

catalog.us-east-1.prod.workshops.aws/event/dashboard/en-US

workshop studio

suhaillariyallur1

### AWS Transform

AWS Transform custom workshop  
— Analyze, Transform, Scale, Measure

Introduction

► Setup and Configuration

► Module 1: Codebase Analysis

▼ AWS account access

[Open AWS console \(us-east-1\)](#)

[Get AWS CLI credentials](#)

Share feedback

Workshop catalog in AWS Builder Center

▼ Content preferences

Language

English

Exit event

### Event outputs (6)

Find outputs

| Key                      | Value                                                                                                                                                                                                                                                     | Stack name  | Description                                         | Type   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------|--------|
| atx-factory-ApiEndpoint  | <a href="https://8chsjvntr2.execute-api.us-east-1.amazonaws.com/prod">https://8chsjvntr2.execute-api.us-east-1.amazonaws.com/prod</a>                                                                                                                     | atx-factory | Base URL for the ATX REST API (IAM-authenticated).  | Export |
| atx-factory-OutputBucket | atx-output-509399592491-us-east-1                                                                                                                                                                                                                         | atx-factory | S3 bucket for transformation outputs.               | Export |
| atx-factory-SourceBucket | atx-source-509399592491-us-east-1                                                                                                                                                                                                                         | atx-factory | S3 bucket for source code uploads.                  | Export |
| DashboardUrl             | <a href="https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards:name=ATX-Transform-CLI-Dashboard">https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards:name=ATX-Transform-CLI-Dashboard</a> | atx-factory | CloudWatch dashboard URL for monitoring batch jobs. | Output |
| JobQueue                 | arn:aws:batch:us-east-1:509399592491:job-queue/atx-job-queue                                                                                                                                                                                              | atx-factory | Name of the Batch job queue.                        | Output |
| URL                      | <a href="https://d3dw1a2c3vaewn.cloudfront.net/?folder=/workshop&amp;tkn=LJOCv7hY2ZGoLn87">https://d3dw1a2c3vaewn.cloudfront.net/?folder=/workshop&amp;tkn=LJOCv7hY2ZGoLn87</a>                                                                           | code-editor | Code Editor URL                                     | Output |

© 2008 - 2026, Amazon Web Services, Inc. or its affiliates. All rights reserved. [Privacy policy](#) [Terms of use](#) [Cookie preferences](#)

d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

workshop

EXPLORER

WORKSHOP

- emoji-algebra
- qwords
- scientific-processor
- utilities

OUTLINE

TIMELINE

APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
participant:/workshop$ atx --version
Version: 3.17.0
participant:/workshop$ uv --version
uv 0.12.23 (aarch64-unknown-linux-gnu)
participant:/workshop$ python3 -v
import _frozen_importlib # frozen
import _imp # builtin
import _thread # <class '_frozen_importlib.BuiltinImporter'>
import _warnings # <class '_frozen_importlib.BuiltinImporter'>
import _weakref # <class '_frozen_importlib.BuiltinImporter'>
import _io # <class '_frozen_importlib.BuiltinImporter'>
import marshal # <class '_frozen_importlib.BuiltinImporter'>
import posix # <class '_frozen_importlib.BuiltinImporter'>
import _frozen_importlib_external # <class '_frozen_importlib.FrozenImporter'>
```

Layout: U.S.



The screenshot shows a VS Code web interface with a terminal window open. The terminal is running a bash shell and executing a series of AWS CLI commands to list and inspect IAM policies. The Explorer sidebar on the left shows a project structure with folders like 'emoji-algebra', 'qwords', 'scientific-processor', and 'utilities'. The terminal output shows the following commands and their results:

```
participant:/workshop$ aws iam list-attached-role-policies --role-name WSParticipantRole
participant:/workshop$ for ARN in \
arn:aws:iam::509399592491:policy/ws-default-policy \
arn:aws:iam::509399592491:policy/standard-workshop-0 \
arn:aws:iam::509399592491:policy/atx-factory-participant-1
do
echo "==== $ARN ===="

VERSION=$(aws iam get-policy \
--policy-arn "$ARN" \
--query 'Policy.DefaultVersionId' \
--output text)

aws iam get-policy-version \
--policy-arn "$ARN" \
--version-id "$VERSION" \
--query 'PolicyVersion.Document' \
--output json
done
==== arn:aws:iam::509399592491:policy/ws-default-policy ====
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowIamReadOnly",
      "Effect": "Allow",
      "Action": [
        "iam:List*",
        "iam:Get*",
        "iam:Generate*",

```

The screenshot shows a VS Code web interface with a terminal window open. The terminal is running a bash shell and executing a series of commands to clone a repository and check prerequisites. The Explorer sidebar on the left shows a project structure with folders like 'aws-transform-custom-samples', 'emoji-algebra', 'qwords', 'scientific-processor', and 'utilities'. The terminal output shows the following commands and their results:

```
participant:/workshop/scientific-processors$ cd ..
participant:/workshop$ git clone https://github.com/aws-samples/aws-transform-custom-samples.git
cd aws-transform-custom-samples/scaled-execution-containers

export AWS_ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
export CDK_DEFAULT_ACCOUNT=$AWS_ACCOUNT_ID
export CDK_DEFAULT_REGION=us-east-1

cd deployment
chmod +x *.sh
./check-prereqs.sh
Cloning into 'aws-transform-custom-samples'...
remote: Enumerating objects: 1670, done.
remote: Counting objects: 100% (308/308), done.
remote: Compressing objects: 100% (161/161), done.
remote: Total 1670 (delta 193), reused 147 (delta 147), pack-reused 1362 (from 1)
Receiving objects: 100% (1670/1670), 1.90 MiB | 8.91 MiB/s, done.
Resolving deltas: 100% (934/934), done.
Prerequisites Check
=====
1. Checking Docker...
   [x] Docker installed: Docker version 29.8.2, build 7fc2dfff
   [x] Docker daemon is running
2. Checking AWS CLI...
   [x] AWS CLI installed: aws-cli/2.37.9 Python/3.14.6 Linux/7.0.0-1013-aws exe/aarch64.ubuntu.24
   [x] AWS CLI v2 detected
   [x] AWS credentials configured
```

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

Explorer sidebar: WORKSHOP, emoji-algebra, qwords, scientific-processor, utilities

Terminal output:

```
aws-transform-custom-samples emoji-algebra qwords scientific-processor utilities
participant:/workshop$ rm -rf aws-transform-custom-samples/
participant:/workshop$ cd /workshop/qwords
rm -rf .venv
uv venv --python 3.8
source .venv/bin/activate
uv pip install -r requirements.txt
python -m pytest tests/ -v
deactivate
Using CPython 3.8.20
Creating virtual environment at: .venv
Activate with: source .venv/bin/activate
Resolved 7 packages in 88ms
Prepared 7 packages in 17ms
Installed 7 packages in 5ms
+ attrs==25.3.0
+ iniconfig==2.1.0
+ packaging==26.2
+ pluggy==1.5.0
+ py==1.11.0
+ pytest==6.2.5
+ tomli==0.10.2

===== test session starts =====
platform linux -- Python 3.8.20, pytest-6.2.5, py-1.11.0, pluggy-1.5.0 -- /workshop/qwords/.venv/bin/python
cachedir: .pytest_cache
rootdir: /workshop/qwords
collected 53 items

tests/test_cli_interface.py::TestDisplayFunctions::test_format_guess_display_all_correct PASSED [ 1%]
tests/test_cli_interface.py::TestDisplayFunctions::test_format_guess_display_mixed_feedback PASSED [ 3%]
```

Status bar: emoji-algebra main 0 AWS: ec2:instance

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

Explorer sidebar: WORKSHOP, emoji-algebra, qwords, scientific-processor, utilities

Terminal output:

```
uv pip install -r requirements.txt -r tests/requirements.txt
python -m pytest tests/ -v
participant:/workshop/scientific-processor$ cd /workshop/scientific-processor

atx custom def exec \
-n AWS/comprehensive-codebase-analysis \
-p . \
-x \
-t

AWS Transform · us-east-1 · AWS/comprehensive-codebase-analysis

Conversation log: /home/participant/.aws/atx/custom/20261005_114048_ff29cbf5/logs/2026-10-05T11-40-49-conversation.log
Monitor progress: tail -f /home/participant/.aws/atx/custom/20261005_114048_ff29cbf5/logs/2026-10-05T11-40-49-conversation.log
Thinking... (8s)
```

Status bar: emoji-algebra main 0 AWS: ec2:instance

scientific-processor > ATXDocumentation > architecture > system-overview.md

# # System Overview

## ## Architecture Summary

The scientific-processor repository implements a **serverless event-driven architecture** using AWS Lambda as the compute layer. The system is designed for direct Lambda invocation (no API Gateway) and performs two categories of operations:

- Scientific data processing** – Statistical analysis, normalization, and percentile calculations using NumPy
- S3 read operations** – Listing buckets, listing objects, and retrieving object metadata

## ## Technology Stack

| Component           | Technology                                    |
|---------------------|-----------------------------------------------|
| Compute             | AWS Lambda (Python 3.9, 128MB, 30s timeout)   |
| IaC                 | AWS SAM (CloudFormation Transform 2016-10-31) |
| Storage Integration | Amazon S3 (read-only)                         |
| Scientific Library  | NumPy 1.21.6                                  |
| AWS SDK             | boto3 (latest, provided by Lambda runtime)    |
| CI/CD               | AWS CodeBuild                                 |
| Architecture        | x86_64 (configurable to arm64)                |

## ## Deployment Model

**Single Lambda function:** `ScientificProcessorFunction`

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
bash - scientific-processor
```

```
5 + ## 1. Command Dispatcher / Strategy Router Pattern
6 + **Location**: 'src/api/processor.py', 'lambda_handler()'
7 +
8 + The Lambda handler uses an 'operation_type' field from the event payload to route to different processing functions. T
9 +
10 +
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

scientific-processor > ATXDocumentation > technical-debt-report.md

# # Technical Debt Report

## ## Critical Issues (Prioritized)

### ## 1. EOL/Deprecated Runtimes – Severity: High

| Component                      | Current Version | Status                        | Risk                |
|--------------------------------|-----------------|-------------------------------|---------------------|
| Python (emoji-algebra, qwords) | 3.8             | <b>End-of-Life</b> (Oct 2024) | No security patches |
| Python (main Lambda)           | 3.9             | <b>End-of-Life</b> (Oct 2025) | Approaching EOL     |

### ## 2. Outdated Dependencies – Severity: Medium

| Dependency | Current Version | Latest Stable | Used By               |
|------------|-----------------|---------------|-----------------------|
| numpy      | 1.21.6          | 2.1.x         | Main project          |
| moto       | 3.1.19          | 5.x           | Test suite            |
| pytest     | 6.2.5           | 8.x           | emoji-algebra, qwords |

### ## 3. Code Quality and Architectural Issues – Severity: Low

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
participant:/workshop$ cd /workshop/qwords
participant:/workshop/qwords$
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

File Explorer (Left):

- WORKSHOP
  - scientific-processor
    - ATXDocumentation
      - technical-debt
        - project-overview.md
        - README.md
        - technical-debt-report.md
      - emoji-algebra
      - events
      - qwords
      - scientific-processor
        - src
        - tests
        - utilities
      - gitignore
      - buildspec.yml
      - env.json
      - LICENSE.md
      - README.md
      - requirements.txt
      - samconfig.toml
      - template.yaml
      - workshop.zip
    - OUTLINE
    - TIMELINE
    - APPLICATION BUILDER

Editor (Right):

scientific-processor > ATXDocumentation > technical-debt-report.md > # Technical Debt Report

```
1 # Technical Debt Report
14
15 ## Critical Issues (Prioritized)
16
17 ### 1. EOL/Deprecated Runtimes - Severity: High
18 | Component | Current Version | Status | Risk |
19 |-----|-----|-----|-----|
20 | Python (emoji-algebra, qwords) | 3.8 | **End-of-Life** (Oct 2024) | No security patches |
21 | Python (main Lambda) | 3.9 | **End-of-Life** (Oct 2025) | Approaching EOL |
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash scient...  
node qwords

**AWS TRANSFORM**

AWS Transform · us-east-1 · AWS/python-version-upgrade

Conversation log:  
Monitor progress: tail -f  
● Read /workshop/qwords (0s)  
● Read /workshop/qwords/requirements.txt (0s)  
● Read /workshop/qwords/app.py (0s)  
● Read /workshop/qwords/tests (0s)  
● Read /workshop/qwords/README.md (0s)  
● Read /workshop/qwords/config.yaml (0s)  
● Read (4 files) (0s)  
● test\_game\_logic.py, test\_models.py, test\_cli\_interface.py, test\_integration.py  
● Git (0s)

scientific-processor atx-result-staging-20261005\_114048\_ff29cbf5\* AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

File Explorer (Left):

- WORKSHOP
  - scientific-processor
    - ATXDocumentation
      - technical-debt
        - project-overview.md
        - README.md
        - technical-debt-report.md
      - emoji-algebra
      - events
      - qwords
      - scientific-processor
        - src
        - tests
        - utilities
      - gitignore
      - buildspec.yml
      - env.json
      - LICENSE.md
      - README.md
      - requirements.txt
      - samconfig.toml
      - template.yaml
      - workshop.zip
    - OUTLINE
    - TIMELINE
    - APPLICATION BUILDER

Editor (Right):

scientific-processor > ATXDocumentation > technical-debt-report.md > # Technical Debt Report

```
1 # Technical Debt Report
14
15 ## Critical Issues (Prioritized)
16
17 ### 1. EOL/Deprecated Runtimes - Severity: High
18 | Component | Current Version | Status | Risk |
19 |-----|-----|-----|-----|
20 | Python (emoji-algebra, qwords) | 3.8 | **End-of-Life** (Oct 2024) | No security patches |
21 | Python (main Lambda) | 3.9 | **End-of-Life** (Oct 2025) | Approaching EOL |
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash scient...  
node qwords  
node emoji...

**AWS TRANSFORM**

AWS Transform · us-east-1 · AWS/python-version-upgrade

Conversation log:  
Monitor progress: tail -f  
● Thinking... (4s)

AWS/python-version-upgrade us-east-1

scientific-processor atx-result-staging-20261005\_114048\_ff29cbf5\* AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

Explorer sidebar: WORKSHOP > scientific-processor > ATXDocumentation > technical-debt > technical-debt-report.md

Technical Debt Report content:

```
1 # Technical Debt Report
14
15 ## Critical Issues (Prioritized)
16
17 ### 1. EOL/Deprecated Runtimes - Severity: High
18 | Component | Current Version | Status | Risk |
```

Terminal output:

```
participant:/workshop$ cd /workshop/emoji-algebra
participant:/workshop/emoji-algebra$ cat > config.yaml << 'EOF'
additionalPlanContext: |
  Target Python 3.12. Modernize all syntax from 3.8 to 3.12.
  Use uv for Python environment management.
  Ensure all existing tests continue to pass.
EOF
participant:/workshop/emoji-algebra$ atx custom def exec \
-n AWS/python-version-upgrade \
-p . \
-c "python -m pytest tests/ -v" \
-g file://config.yaml \
-x \
-t
```

AWS TRANSFORM logo

AWS Transform · us-east-1 · AWS/python-version-upgrade

Conversation log: Monitor progress: tail -f

scientific-processor atx-result-staging-20261005\_114048\_ff29cbf5 AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

Browser address bar: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

Explorer sidebar: WORKSHOP > scientific-processor > ATXDocumentation > technical-debt > technical-debt-report.md

Technical Debt Report content:

```
1 # Technical Debt Report
14
15 ## Critical Issues (Prioritized)
16
17 ### 1. EOL/Deprecated Runtimes - Severity: High
18 | Component | Current Version | Status | Risk |
```

Terminal output:

```
participant:/workshop$ # Find active session IDs
ls ~/.aws/atx/custom/

# Tail the conversation log (path shown in ATX output)
tail -f ~/.aws/atx/custom/<SESSION-ID>/logs/*_log
20261005_114048_ff29cbf5 20261005_120800_6047fe1a 20261005_121021_cd6e0715
bash: SESSION-ID: No such file or directory
participant:/workshop$ tail -f ~/.aws/atx/custom/20261005_120800_6047fe1a/logs/*_log
2. **All 53 unit tests pass** - 100% pass rate maintained on Python 3.12
3. **Dependencies compatible** - pytest upgraded from 6.2.5 to 9.0.3 for Python 3.12 compatibility
4. **Documentation updated** - README.md reflects Python 3.12, uv package manager, and updated pytest version
5. **Deployment scripts** - Not applicable (none exist)

Key changes made:
- **app.py**: Converted all `.format()` calls to f-strings; updated version references to 3.12
- **requirements.txt**: pytest 6.2.5 → 9.0.3
- **README.md**: Python 3.8 → 3.12 references, added uv as prerequisite, updated instructions
=== Agent Minutes used: 45.65 ===== Conversation Ended: 2026-10-05T12:11:34.969Z ===

^C
participant:/workshop$ ls ~/.aws/atx/custom/
20261005_114048_ff29cbf5 20261005_120800_6047fe1a 20261005_121021_cd6e0715
participant:/workshop$ cat ~/.aws/atx/custom/20261005_121021_cd6e0715/artifacts/worklog.log
cat: /home/participant/.aws/atx/custom/20261005_121021_cd6e0715/artifacts/worklog.log: No such file or director
y
participant:/workshop$ cat ~/.aws/atx/custom/20261005_114048_ff29cbf5/artifacts/worklog.log
cat: /home/participant/.aws/atx/custom/20261005_114048_ff29cbf5/artifacts/worklog.log: No such file or director
y
participant:/workshop$
```

scientific-processor atx-result-staging-20261005\_114048\_ff29cbf5 AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py

app.py

```
1 #!/usr/bin/env python3
2
3 QWords - A Wordle-like word guessing game
4 Python 3.12 implementation for SEG transformation demo
5
6
7 import random
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
1 # Validation Result Summary
2
3 ## TRANSFORMATION SUMMARY
4 BEFORE: Python 3.8 application (emoji-algebra puzzle game) with a single source file 'app.py' using `
5
6 AFTER: The transformation migrated the project to Python 3.12. Changes made:
7 - 'app.py': Converted all 34 `.format()` calls to f-strings; converted 4 if/elif/else chains to match
8 - 'requirements.txt': Updated `pytest==6.2.5` to `pytest==9.0.3`.
9 - 'README.md': Updated all Python version references from 3.8 to 3.12, updated venv setup instruction
10 - All 48 existing tests continue to pass on Python 3.12.3.
11
12 ## OVERALL STATUS: COMPLETE
13
14 ## EXIT CRITERIA RESULTS
15 Read ~/.aws/ats/custom/20261005_121021_cd6e0715/artifacts/validation_summary.md (0s)
16 Read ~/.aws/ats/custom/20261005_121021_cd6e0715/artifacts/worklog.log (0s)
17
18 ✓ Transformation complete
19
20 Files changed: 4
21 Lines added: 197
22 Lines deleted: 203
23 Lines modified: 400
24 Initial commit: b90a2451
25
26 - README.md
27 - app.py
28 - config.yaml
29 - requirements.txt
```

AWS/python-version-upgrade us-east-1

participant:/workshop/emoji-algebra

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

```
participant:/workshop/emoji-algebra$ cd /workshop/qwords
git log --oneline
b4de068 (HEAD -> atx-result-staging-20261005_120800_6047fe1a) Step 1: Migrate Python 3.8 to 3.12 - upgrade pyte
st 6.2.5 to 9.0.3, convert .format() to f-strings, update README for Python 3.12 and uv. Build status: Success
(53/53 tests passed)
2d72d71 (main) Initial commit
participant:/workshop/qwords$
```

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
((.venv) ) participant:/workshop/scientific-processor$ atx
AWS
TRANSFORM
AWS Transform · us-east-1
You
I want to create a reusable transformation definition that upgrades Python 3.9
Lambda functions to 3.12, replaces raw boto3 S3 calls with a repository pattern,
adds structured logging with aws-lambda-powertools, and adds input validation
with Pydantic.
Use /workshop/scientific-processor as the source project to author and test
against. The ATXDocumentation/ folder in that project contains architecture
and dependency analysis - use it as reference context for understanding the
code patterns.
Use "python3.12 -m pytest tests/ -v" as the validation command.
Conversation log:
/home/participant/.aws/ats/custom/20261005_134335_bfe57c7c/logs/2026-10-05T13-43-37-conversation.log
Monitor progress: tail -f
/home/participant/.aws/ats/custom/20261005_134335_bfe57c7c/logs/2026-10-05T13-43-37-conversation.log
Resume: atx --conversation-id 20261005_134335_bfe57c7c
Thinking... (5s)
us-east-1
type a message or / for commands
```

qwords atx-result-staging-20261005\_120800\_6047fe1a

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
      - invoke-api.py
- OUTLINE
- TIMELINE
- APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
((.venv) ) participant:/workshop/scientific-processor$ atx
and dependency analysis - use it as reference context for understanding the
code patterns.

Use "python3.12 -m pytest tests/ -v" as the validation command.
Conversation log:
/home/participant/.aws/atx/custom/20261005_134335_bfe57c7c/logs/2026-10-05T13-43-37-conversation.log
Monitor progress: tail -f
/home/participant/.aws/atx/custom/20261005_134335_bfe57c7c/logs/2026-10-05T13-43-37-conversation.log
Resume: atx --conversation-id 20261005_134335_bfe57c7c

| ATX
Let me start by checking for any existing transformations that might cover this use case and
consulting the AWS Transform portfolio.
• List Transformation Definitions (0s)
  Found 37 transformation definitions
    - 37 AWS Managed transformations

| ATX
I found one relevant AWS managed transformation -- "AWS/python-version-upgrade" -- which handles
Python 3.8/3.9 to 3.12/3.13 upgrades. However, your request goes well beyond a version bump. You
also want a repository pattern for S3, structured logging with aws-lambda-powertools, and Pydantic
validation. A custom transformation is the right approach here since it can address all four
concerns together.

No existing user transformations were found, so let me kick off the requirements interview. I'll
first scan your project and the ATX documentation folder to understand the codebase, then ask you a
few targeted questions to nail down the details. Good scope - Python 3.9 to 3.12 combined with
architectural improvements (repository pattern, Powertools logging, Pydantic validation) is a solid
modernization package. Let me scan your project and the documentation folder to understand what
we're working with.
• Handoff EXPLORE
  EXPLORE, Scan /workshop/scientific-processor thoroughly. I need:
    1...
  • Handoff EXPLORE
    EXPLORE, Read all files in /workshop/scientific-processor/ATXDocum...
  • Analyzing full-scan-results.part-002.md... (3s)

us-east-1 ~18.15m
Agent is working · type to queue a message
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
      - invoke-api.py
- OUTLINE
- TIMELINE
- APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
((.venv) ) participant:/workshop/scientific-processor$ atx
Great, I've got a solid picture of your project. Here's what I found:

• Single Lambda handler (src/api/processor.py) dispatching on operation_type: health, process,
s3_list_buckets, s3_list_objects, s3_get_object
• Three raw boto3 S3 calls (list_buckets, list_objects_v2, head_object) -- all read-only, no
pagination or retry logic
• Zero logging anywhere -- errors silently returned as dicts
• Minimal validation -- just presence checks on required fields, no type/format validation
• pytest with unittest.TestCase, only 5 tests, several gaps
• SAM template targeting python3.9, numpy 1.21.6, moto 3.1.19

Here are some decision questions to shape the transformation:

1. S3 Repository Pattern -- Should the repository class be a generic interface (abstract base
class with S3 as one implementation), or a concrete S3Repository class purpose-built for this
project's read-only operations (list_buckets, list_objects, head_object)? My recommendation:
Concrete S3Repository class since you only have read-only operations and a single storage backend.
Keep it simple.

2. Powertools Logging Scope -- Should we add logging at just the handler entry/exit points, or
instrument every operation (each S3 call, each data processing step, error paths)? My
recommendation: Instrument comprehensively -- log handler entry with correlation ID, each S3
operation with timing, data processing steps, and all error paths with structured context.
Powertools makes this low-effort.

3. Pydantic Validation Models -- Should we create one Pydantic model per operation type (e.g.,
S3ListObjectsRequest, ProcessRequest) or a single event model with optional fields and a
discriminator? My recommendation: One model per operation type, with a base event model and a
discriminated union on operation_type. This gives you clear error messages per operation.

4. Dependency Updates -- numpy is pinned at 1.21.6 (doesn't support 3.12) and moto at 3.1.19 (very
old). Should the transformation also bump these to 3.12-compatible versions? My recommendation: Yes,
update numpy to >=1.26 and moto to >=5.0 as part of the upgrade since they won't work on 3.12
otherwise.

You can say "use defaults" to accept the recommendations and continue, or "done" to end the
interview and start creating.

us-east-1
use defaults
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

((.venv) ) participant:/workshop/scientific-processor\$ atx

discriminated union on operation\_type. This gives you clear error messages per operation.

4. Dependency Updates — numpy is pinned at 1.21.6 (doesn't support 3.12) and moto at 3.1.19 (very old). Should the transformation also bump these to 3.12-compatible versions? My recommendation: Yes, update numpy to >=1.26 and moto to >=5.0 as part of the upgrade since they won't work on 3.12 otherwise.

You can say "use defaults" to accept the recommendations and continue, or "done" to end the interview and start creating.

You use defaults

ATX Got it, defaults accepted for all four. A few more questions:

5. Error Handling — Currently all errors are caught as bare 'except Exception' with no categorization. Should the transformation introduce typed exceptions (e.g., S3ObjectNotFoundError, ValidationError, ProcessingError) with appropriate HTTP-style status codes in responses? My recommendation: Yes, create a small exception hierarchy and map them to structured error responses with error codes, so callers can distinguish between validation failures, not-found, and internal errors.

6. Powertools Extras — Beyond structured logging, aws-lambda-powertools also offers tracing (X-Ray) and metrics (CloudWatch EMF). Should we add those too, or keep it to logging only? My recommendation: Add tracing as well since it's minimal effort with the @tracer.capture\_lambda\_handler decorator and gives you X-Ray visibility into S3 calls. Skip metrics for now to keep scope focused.

7. Test Modernization — The existing tests use unittest.TestCase style with mock.patch. Should the transformation rewrite them to pure pytest style (fixtures, parametrize, no class inheritance) and add coverage for the currently untested operations (s3\_list\_objects, s3\_get\_object, normalize, percentiles)? My recommendation: Yes, convert to pure pytest style and expand coverage to all operations, including the new repository and validation layers.

You can say "use defaults" to accept the recommendations and continue, or "done" to end the interview and start creating.

us-east-1  
type a message or / for commands

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

((.venv) ) participant:/workshop/scientific-processor\$ atx

@tracer.capture\_lambda\_handler decorator and gives you X-Ray visibility into S3 calls. Skip metrics for now to keep scope focused.

7. Test Modernization — The existing tests use unittest.TestCase style with mock.patch. Should the transformation rewrite them to pure pytest style (fixtures, parametrize, no class inheritance) and add coverage for the currently untested operations (s3\_list\_objects, s3\_get\_object, normalize, percentiles)? My recommendation: Yes, convert to pure pytest style and expand coverage to all operations, including the new repository and validation layers.

You can say "use defaults" to accept the recommendations and continue, or "done" to end the interview and start creating.

You use defaults

ATX Defaults accepted. Two final questions:

8. SAM Template Updates — The template.yaml needs the runtime changed to python3.12. Should the transformation also add a Powertools Lambda layer ARN and update the IAM policy to scope the S3 resource from "\*" to a parameterized bucket ARN? My recommendation: Yes to both. Use the AWS-managed Powertools layer for python3.12, and add a SAM parameter for the bucket name to tighten the IAM policy.

9. Nested Duplicate Directory — There's a full copy of the project at 'scientific-processor/scientific-processor/'. Should the transformation ignore it entirely, or flag/remove it? My recommendation: Ignore it — the transformation should only target 'src/', 'tests/', 'template.yaml', 'requirements.txt', and 'buildspec.yml'. Cleanup of the duplicate is a separate concern.

You can say "use defaults" to accept the recommendations and continue, or "done" to end the interview and start creating.

You use defaults

Thinking... (11s)  
This is taking a moment...

us-east-1 ~27.05m  
Agent is working · type to queue a message

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

9 + Consult 'references/architecture/' for component details, dependency maps, and design-pattern context  
10 +  
11 +  
12 + ## Entry Criteria  
13 +  
14 + 1. Project is a Python 3.9 AWS Lambda function deployed via AWS SAM (template.yaml with 'Runtime: py  
15 + 2. Source code contains direct boto3 S3 client calls (e.g., 'boto3.client('s3')' with operations suc  
16 + 3. Project does not already use aws-lambda-powertools for logging or tracing.  
17 + 4. Project does not already use Pydantic for request/event validation.  
18 + 5. A 'requirements.txt' (or equivalent dependency file) exists.  
19 + 6. A test suite exists using pytest.  
20 +  
... 133 more

ATX  
The transformation definition has been created. Here's what was built:

Name: python39-lambda-to-312-modernized  
Location: ~/.aws/atx/custom/20261005\_134335\_bfe57c7c/artifacts/tp-staging/SKILL.md

It covers 8 phases across 22 implementation steps:

- Phase 1: Python 3.12 version bump, dependency updates (numpy >=1.26, moto >=5.0), SAM template runtime + Powertools layer + parameterized bucket ARN
- Phase 2: Typed exception hierarchy (ServiceError, ValidationError, S3ObjectNotFoundError, S3AccessDeniedError, ProcessingError)
- Phase 3: Pydantic v2 models — one per operation type with a discriminated parse\_event dispatcher
- Phase 4: Concrete S3Repository class with Powertools logging and tracing in every method
- Phase 5: Handler refactor — removes raw boto3, wires up repository/validation/logging/tracing
- Phase 6: X-Ray tracing enabled in SAM template
- Phase 7: Full test modernization — pure pytest style, coverage for all operations, models, and repository error paths
- Phase 8: Syntax and import verification checks

The ATXDocumentation reference docs (architecture, components, dependencies, patterns) were copied into the references/ folder so the execution agent can consult them at runtime.

Would you like me to review the quality of this transformation definition before we move forward?

us-east-1  
type a message or / for commands

atx-result-staging-20261005\_120800\_6047fe1a 0 0 0 0 AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

9 + "template.yaml",  
10 + "requirements.txt",  
11 + "tests/requirements.txt",  
12 + "buildspec.yml"  
13 + },  
14 + "details": {  
15 + "template.yaml: Runtime python3.9 -> python3.12, add BucketName parameter, scope IAM to \${Bu  
16 + "requirements.txt: numpy>=1.26,<2.0, boto3, aws-lambda-powertools[tracer]>=3.0, pydantic>=2.  
17 + "tests/requirements.txt: pytest>=7, moto>=5.0, requests, pydantic>=2.5",  
18 + "buildspec.yml: python 3.9 -> 3.12"  
19 + }  
20 + },  
... 85 more

ATX  
Here's the transformation plan summary:

7 Steps to transform scientific-processor from Python 3.9 to 3.12 with modernization:

1. **Config updates** - template.yaml (runtime 3.12, Powertools layer, scoped IAM, tracing), requirements.txt (numpy>=1.26, powertools, pydantic), buildspec.yml
2. **Exception hierarchy** - New 'src/api/exceptions.py' with ServiceError base + typed subclasses (ValidationError, S3ObjectNotFoundError, S3AccessDeniedError, ProcessingError)
3. **Pydantic models** - New 'src/api/models.py' with per-operation validation models and a 'parse\_event()' dispatcher
4. **S3 Repository** - New 'src/api/s3\_repository.py' encapsulating all S3 operations with Powertools logging/tracing and typed error mapping
5. **Refactor handler** - Update 'src/api/processor.py' to use S3Repository, Powertools Logger/Tracer, Pydantic validation, typed exceptions
6. **Modernize tests** - Pure pytest style (no unittest.TestCase), new test files for models and S3 repository, moto v5 '@mock\_aws', comprehensive coverage
7. **Verify** - py\_compile all files + full test suite

All target dependency versions passed CVE checks.

The full plan is at '~/.aws/atx/custom/20261005\_134335\_bfe57c7c/artifacts/plan.json'. Would you like to proceed with the plan, or do you have modifications?

us-east-1  
type a message or / for commands

atx-result-staging-20261005\_120800\_6047fe1a 0 0 0 0 AWS: ec2:instance Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py
  - OUTLINE
  - TIMELINE
  - APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ATX

Now add the TRANSFORMATION SUMMARY and OVERALL STATUS at the top:

- Edit validation\_summary.md (0s)

```
1 # Validation Result Summary
2
3 ## TRANSFORMATION SUMMARY
4 + BEFORE: Python 3.9 Lambda function deployed via SAM with `Runtime: python3.9`, direct `boto3.client`
5 +
6 + AFTER: Python 3.12 Lambda with `Runtime: python3.12`, Powertools layer, `Tracing: Active`, parameteri
7 +
8 ## OVERALL STATUS: COMPLETE
9 +
10 ## EXIT CRITERIA RESULTS
```

ATX

Validation is complete. All 10 exit criteria pass.

- Read ~/.aws/ats/custom/20261005\_134335\_bfe57c7c/artifacts/validation\_summary.md (0s)

ATX

The transformation is complete. Here's a summary of the results:

All 10 exit criteria passed:

- All 39 pytest tests pass on Python 3.12
- 'template.yaml' updated with Python 3.12 runtime, Powertools layer, active tracing, and parameterized bucket ARN
- 'requirements.txt' includes 'numpy>=1.26,<2.0', 'aws-lambda-powertools[tracer]>=3.0', and 'pydantic>=2.5'
- No direct 'boto3.client('s3')' calls in handler — all S3 ops go through 'S3Repository'
- All Lambda events validated through Pydantic models before processing
- All logging uses Powertools 'Logger' — no 'print()' or raw 'logging'
- All errors return structured responses via typed exception hierarchy
- X-Ray tracing decorators on handler and all S3 repository methods
- Tests use pure pytest style with full coverage of operations and error paths
- All '.py' files compile cleanly under Python 3.12

Do you have any questions or feedback about the transformation?

us-east-1

type a message or / for commands ↵

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py
  - OUTLINE
  - TIMELINE
  - APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

3. Apply it to a different codebase  
4. Modify the transformation definition based on what you observed  
5. Scan the repository for other applicable transformations

You

publish registry

ATX

Let me pull the name and description from the transformation definition.

Name: python39-lambda-to-312-modernized  
Description: Upgrade Python 3.9 Lambda functions to 3.12 with S3 repository pattern, aws-lambda-powertools structured logging and tracing, Pydantic input validation, typed exceptions, and modernized pytest test suite.

Would you like to use these as-is, or modify either the name or description before publishing?

You

as-is

- Save Transformation (1s)

```
python39-lambda-to-312-modernized, Upgrade Python 3.9 Lambda functions to 3.12 with S3 repos..., publish: true, is_new_transformation: false
```

ATX

Published successfully.

Name: python39-lambda-to-312-modernized  
Version: 3gzsxh26h0md7fhendlfxp66

The transformation is now discoverable by other users in your account. What would you like to do next?

- Apply it to another codebase
- Create a new transformation
- Modify this transformation
- List available transformations

us-east-1

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - scientific-processor
    - utilities
    - invoke-api.py
  - OUTLINE
  - TIMELINE
  - APPLICATION BUILDER

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
((.venv) ) participant:/workshop/scientific-processor$ cd /workshop/scientific-processor

# Switch to the latest ATX result branch (if not already on it)
LATEST_BRANCH=$(git branch | grep atx-result | tail -1 | tr -d ' ')
git checkout "$LATEST_BRANCH"

python3.12 -m venv .venv --clear
source .venv/bin/activate
pip install -r requirements.txt --r tests/requirements.txt -q
python3.12 -m pytest tests/ -v
Already on 'atx-result-staging-20261005_134335_bfe57c7c'

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip

===== test session starts =====
platform linux -- Python 3.12.15, pytest-9.1.1, pluggy-1.6.0 -- /workshop/scientific-processor/.venv/bin/python
cachedir: .pytest_cache
rootdir: /workshop/scientific-processor
collected 39 items

tests/unit/test_models.py::test_health_check_request_valid PASSED [ 2%]
tests/unit/test_models.py::test_process_request_valid PASSED [ 5%]
tests/unit/test_models.py::test_process_request_empty_data PASSED [ 7%]
tests/unit/test_models.py::test_process_request_invalid_data PASSED [ 10%]
tests/unit/test_models.py::test_s3_list_buckets_request_valid PASSED [ 12%]
tests/unit/test_models.py::test_s3_list_objects_request_valid PASSED [ 15%]
tests/unit/test_models.py::test_s3_list_objects_request_short_bucket PASSED [ 17%]
tests/unit/test_models.py::test_s3_list_objects_request_max_keys_bounds PASSED [ 20%]
tests/unit/test_models.py::test_s3_get_object_request_valid PASSED [ 23%]
tests/unit/test_models.py::test_s3_get_object_request_empty_key PASSED [ 25%]
tests/unit/test_models.py::test_parse_event_health PASSED [ 28%]
tests/unit/test_models.py::test_parse_event_process PASSED [ 30%]
tests/unit/test_models.py::test_parse_event_s3_list_buckets PASSED [ 33%]
tests/unit/test_models.py::test_parse_event_s3_list_objects PASSED [ 35%]
tests/unit/test_models.py::test_parse_event_s3_get_object PASSED [ 38%]
tests/unit/test_models.py::test_parse_event_missing_operation_type PASSED [ 41%]
tests/unit/test_models.py::test_parse_event_unsupported_operation_type PASSED [ 43%]
tests/unit/test_models.py::test_parse_event_invalid_data PASSED [ 46%]
tests/unit/test_processor.py::test_health_check PASSED [ 48%]
tests/unit/test_processor.py::test_process_valid_data PASSED [ 51%]
tests/unit/test_processor.py::test_process_normalize PASSED [ 53%]
tests/unit/test_processor.py::test_process_percentiles PASSED [ 56%]
tests/unit/test_processor.py::test_process_empty_data PASSED [ 58%]
```

workshop

SOURCE CONTROL

CHANGES

- emoji... atx-result-staging ✓ ↻ ...  
Message (%Enter to commit on "atx-...")  
Publish Branch
- qwor... atx-result-staging ✓ ↻ ...  
Message (%Enter to commit on "atx-...")  
Publish Branch
- scient... atx-result-staging ✓ ↻ ...  
Message (%Enter to commit on "atx-...")  
Publish Branch

Step 1: Migrate from... atx-result-stagi...

Initial commit Workshop Participant

- .gitignore emoji-algebra A
- LICENSE.md emoji-algebra A
- README.md emoji-algebra A
- app.py emoji-algebra A
- requirements.txt emoji-algebra A
- \_\_init\_\_.py emoji-algebra/tests A
- test\_cli\_interface.py emoji-algebra/tests A
- test\_game\_logic.py emoji-algebra/tests A
- test\_integration.py emoji-algebra/tests A
- test\_models.py emoji-algebra/tests A

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
((.venv) ) participant:/workshop/scientific-processor$ cd /workshop/scientific-processor

# Switch to the latest ATX result branch (if not already on it)
LATEST_BRANCH=$(git branch | grep atx-result | tail -1 | tr -d ' ')
git checkout "$LATEST_BRANCH"

tests/unit/test_models.py::test_process_request_invalid_data PASSED [ 10%]
tests/unit/test_models.py::test_s3_list_buckets_request_valid PASSED [ 12%]
tests/unit/test_models.py::test_s3_list_objects_request_valid PASSED [ 15%]
tests/unit/test_models.py::test_s3_list_objects_request_short_bucket PASSED [ 17%]
tests/unit/test_models.py::test_s3_list_objects_request_max_keys_bounds PASSED [ 20%]
tests/unit/test_models.py::test_s3_get_object_request_valid PASSED [ 23%]
tests/unit/test_models.py::test_s3_get_object_request_empty_key PASSED [ 25%]
tests/unit/test_models.py::test_parse_event_health PASSED [ 28%]
tests/unit/test_models.py::test_parse_event_process PASSED [ 30%]
tests/unit/test_models.py::test_parse_event_s3_list_buckets PASSED [ 33%]
tests/unit/test_models.py::test_parse_event_s3_list_objects PASSED [ 35%]
tests/unit/test_models.py::test_parse_event_s3_get_object PASSED [ 38%]
tests/unit/test_models.py::test_parse_event_missing_operation_type PASSED [ 41%]
tests/unit/test_models.py::test_parse_event_unsupported_operation_type PASSED [ 43%]
tests/unit/test_processor.py::test_health_check PASSED [ 48%]
tests/unit/test_processor.py::test_process_valid_data PASSED [ 51%]
tests/unit/test_processor.py::test_process_normalize PASSED [ 53%]
tests/unit/test_processor.py::test_process_percentiles PASSED [ 56%]
tests/unit/test_processor.py::test_process_empty_data PASSED [ 58%]
tests/unit/test_processor.py::test_process_invalid_data_type PASSED [ 61%]
tests/unit/test_processor.py::test_process_data_stats_direct PASSED [ 64%]
tests/unit/test_processor.py::test_process_invalid_operation PASSED [ 66%]
tests/unit/test_processor.py::test_s3_list_buckets PASSED [ 69%]
tests/unit/test_processor.py::test_s3_list_objects PASSED [ 71%]
tests/unit/test_processor.py::test_s3_list_objects_missing_bucket PASSED [ 74%]
tests/unit/test_processor.py::test_s3_get_object PASSED [ 76%]
tests/unit/test_processor.py::test_s3_get_object_not_found PASSED [ 79%]
tests/unit/test_processor.py::test_invalid_operation_type PASSED [ 82%]
tests/unit/test_processor.py::test_missing_operation_type PASSED [ 84%]
tests/unit/test_s3_repository.py::test_list_buckets PASSED [ 87%]
tests/unit/test_s3_repository.py::test_list_objects PASSED [ 89%]
tests/unit/test_s3_repository.py::test_list_objects_empty_bucket PASSED [ 92%]
tests/unit/test_s3_repository.py::test_list_objects_nonexistent_bucket PASSED [ 94%]
tests/unit/test_s3_repository.py::test_get_object_metadata PASSED [ 97%]
tests/unit/test_s3_repository.py::test_get_object_metadata_not_found PASSED [ 100%]

===== 39 passed in 0.82s =====

((.venv) ) participant:/workshop/scientific-processor$
```

workshop

processor.py (5fce8d6) ↔ processor.py (73bd17e)

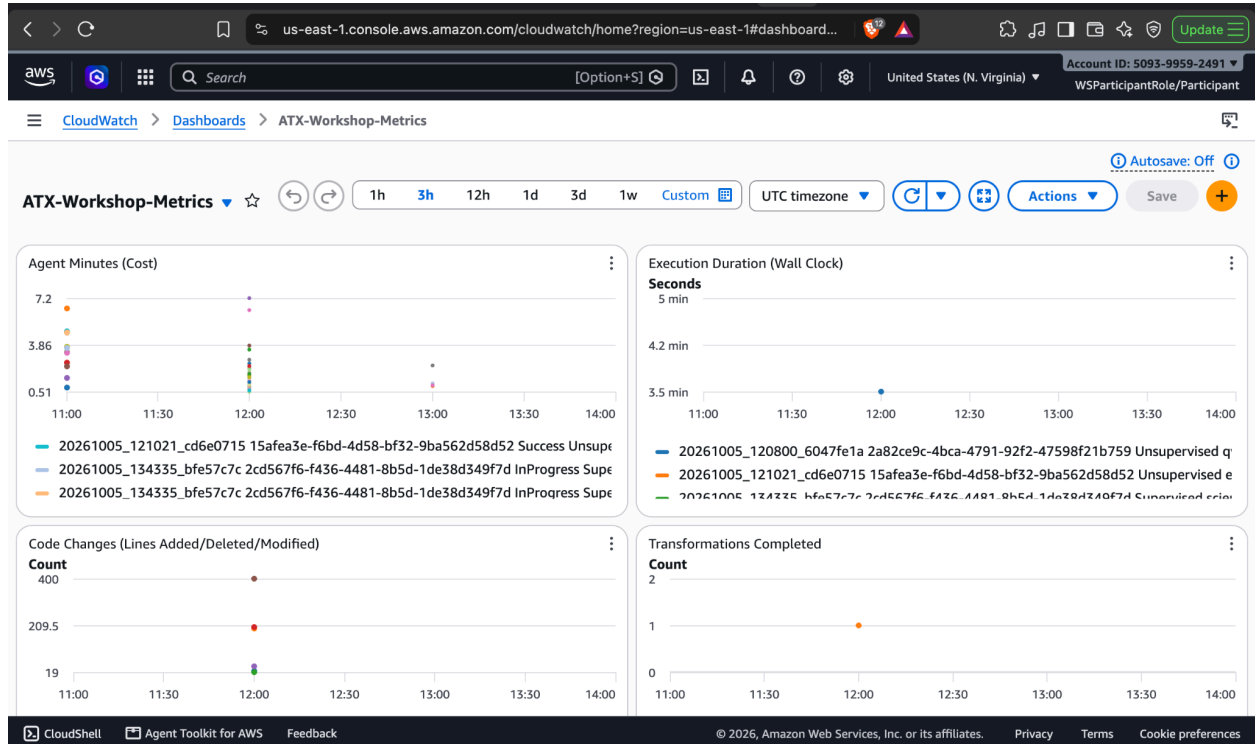
workshop > scientific-processor > src > api > processor.py

```
1 Copyright Amazon.com, Inc. or its affiliates. All Rights Reserved.
2 SPDX-License-Identifier: MIT-0
3
4 import json
5 import sys
6 import boto3
7 import numpy as np
8 from typing import Dict, List, Union
9
10 AWS SDK v1 - S3 client
11 s3_client = boto3.client('s3')
```

worklog.log

```
((.venv) ) participant:/workshop/scientific-processor$ cat ~/.aws/atx/custom/20261005_134335_bfe57c7c/artifacts
/worklog.log
## Worklog

### Entry 1 - Full transformation complete
**What:** Migrated scientific-processor Lambda from Python 3.9 to 3.12 with full architectural modernization.
**Files changed:**
- template.yaml (Runtime 3.12, Powertools layer, Tracing Active, BucketName param, scoped IAM)
- requirements.txt (numpy>=1.26,<2.0, aws-lambda-powertools[tracer]>=3.0, pydantic>=2.5)
- tests/requirements.txt (moto>=5.0, pydantic>=2.5)
- buildspec.yml (python 3.12)
- src/api/exceptions.py (NEW - typed exception hierarchy)
- src/api/models.py (NEW - Pydantic v2 validation models)
- src/api/s3_repository.py (NEW - S3Repository with Powertools logging/tracing)
- src/api/processor.py (refactored - Logger, Tracer, parse_event, S3Repository, structured errors)
- tests/conftest.py (NEW - shared fixtures including FakeLambdaContext)
- tests/unit/test_processor.py (rewritten - pure pytest, 14 tests)
- tests/unit/test_models.py (NEW - 18 Pydantic model tests)
- tests/unit/test_s3_repository.py (NEW - 6 moto-based S3Repository tests)
**Verification:** python3.12 -m pytest tests/ -v - 39 passed; 'py_compile' all files - success
**Issues:** Initial test run failed because Powertools '@logger.inject_lambda_context' requires a proper Lambda
context object, not a plain dict. Fixed by creating FakeLambdaContext class in conftest.py.
((.venv) ) participant:/workshop/scientific-processor$
```





us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log...

CloudWatch > Log management > /aws/batch/atx-transform > atx/default/ca58e13a43d242e0967ef6571541fa83

### CloudWatch

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

1m 1h UTC timezone

Display

| Timestamp                | Message                                                                               |
|--------------------------|---------------------------------------------------------------------------------------|
| 2026-10-05T15:17:38.970Z | There are older events to load. <a href="#">Load more</a> .                           |
| 2026-10-05T15:17:38.970Z | moval (socket pools, response parsers), test infrastructure                           |
| 2026-10-05T15:17:38.970Z | update, and parity testing. Triggers: PayShield, HSM migrati                          |
| 2026-10-05T15:17:38.970Z | on, AWS Payment Cryptography, APC, DUKPT, TranslatePinData,                           |
| 2026-10-05T15:17:38.970Z | DecryptData, VerifyPinData.                                                           |
| 2026-10-05T15:17:38.970Z | 31   AWS/dotnet-lambda-upgrade   Upgrades AWS Lambda .NET functions from older run... |
| 2026-10-05T15:17:38.970Z | Core 3.1, .NET 6, .NET 8) to .NET 10. Covers project files                            |
| 2026-10-05T15:17:38.970Z | (.csproj/.fsproj), NuGet packages (Amazon.Lambda.*, AWSSDK.*                          |
| 2026-10-05T15:17:38.970Z | , Microsoft.Extensions.*), Lambda configuration (SAM, Server                          |
| 2026-10-05T15:17:38.970Z | less Framework, CDK, Terraform, CloudFormation), Dockerfiles                          |

CloudShell Agent Toolkit for AWS Feedback © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

workshop

EXPLORER

- WORKSHOP
  - emoji-algebra
  - qwords
    - \_\_pycache\_\_
    - .pytest\_cache
    - .venv
    - tests
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - results
    - scientific-processor
    - utilities
      - invoke-api.py
- OUTLINE
- TIMELINE
- APPLICATION BUILDER

invoke-api.py

```
1 #!/usr/bin/env python3
2 """
3 Helper script to invoke AWS Transform CLI API with automatic IAM signing.
4
5 Uses boto3 for SigV4 authentication - works with EC2 instance roles,
6 environment variables, SSO profiles, or any standard AWS credential source.
7
8 Usage:
9 python3 utilities/invoke-api.py \
10
11 ({.venv}) participant:/workshop$ python3 utilities/invoke-api.py \
12 --endpoint "$API_ENDPOINT" \
13 --method GET \
14 --path "/jobs/batch/$BATCH_ID"
15 {
16   "STARTING": 0,
17   "RUNNING": 3,
18   "SUCCEEDED": 0,
19   "FAILED": 0
20 }
21
22 ({.venv}) participant:/workshop$ export BATCH_ID="codebase-analysis-2026-20261005-151533"
23 ({.venv}) participant:/workshop$ AWS_DEFAULT_REGION=us-east-1 python3 utilities/invoke-api.py --endpoint "$API_ENDPOINT" --method GET --path "/jobs/batch/$BATCH_ID"
24 {
25   "batchId": "codebase-analysis-2026-20261005-151533",
26   "status": "COMPLETE",
27   "progress": 100.0,
28   "totalJobs": 3,
29   "statusCounts": {
30     "SUBMITTED": 0,
31     "PENDING": 0,
32     "RUNNABLE": 0,
33     "STARTING": 0,
34     "RUNNING": 0,
35     "SUCCEEDED": 3,
36     "FAILED": 0
37   }
38 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash sc... bash q... bash

Ln, Col 1 Spaces: 4 UTF-8 LF {} Python 3.12.15 64-bit Layout: U.S.

Browser: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

workshop

EXPLORER

- WORKSHOP
  - qwords
    - .gitignore
    - app.py
    - config.yaml
    - LICENSE.md
    - README.md
    - requirements.txt
    - results
      - spring-petclinic-comprehensive-c...
        - code/spring-petclinic
          - .devcontainer
          - .github
          - .myn
          - ATXDocumentation
          - gradle
          - k8s
          - db.yml
          - petclinic.yml
          - src
          - .editorconfig
          - .gitattributes
          - .gitignore
          - gitpod.yml
          - build.gradle

db.yml

```
1 ---
2 apiVersion: v1
3 kind: Secret
4 metadata:
5   name: demo-db
6 type: servicebinding.io/postgresql
7 stringData:
8   type: "postgresql"
9   provider: "postgresql"
10  host: "demo-db"
11  port: "5432"
12  database: "petclinic"
13  username: "user"
14  password: "pass"
15 ---
16 apiVersion: v1
17 kind: Service
18 metadata:
19   name: demo-db
20 spec:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

sis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/objects/71/b20e6ba0029c8309405a2b41b281fe541bba33 to r  
esults/toapilambdanode16-comprehensive-codebase-analysis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/o  
bjects/71/b20e6ba0029c8309405a2b41b281fe541bba33  
download: s3://atx-output-509399592491-us-east-1/transformations/toapilambdanode16-comprehensive-codebase-analy  
sis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/objects/97/3a615e73f2a3109f4ee691928749eb86e85bf8 to r  
esults/toapilambdanode16-comprehensive-codebase-analysis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/o  
bjects/97/3a615e73f2a3109f4ee691928749eb86e85bf8  
download: s3://atx-output-509399592491-us-east-1/transformations/toapilambdanode16-comprehensive-codebase-analy  
sis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/objects/97/048f9ebf9966915490730f0ddcf5bca7c8d329 to r  
esults/toapilambdanode16-comprehensive-codebase-analysis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/o  
bjects/97/048f9ebf9966915490730f0ddcf5bca7c8d329  
download: s3://atx-output-509399592491-us-east-1/transformations/toapilambdanode16-comprehensive-codebase-analy  
sis/20261005\_152409\_514169ef/code/toapilambdanode16/.git/objects/95/b4f011c0bf47a466b86ad4a8a115a37b77ba18 to r

Ln 1, Col 1 Spaces: 2 UTF-8 LF ( ) YAML Layout: U.S.

Browser: d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

workshop

EXPLORER

- WORKSHOP
  - results
    - spring-petclinic-comprehensive-c...
      - code/spring-petclinic
        - mvrnw.cmd
        - pom.xml
        - README.md
        - settings.gradle
        - logs
          - custom/20261005\_151702\_ffa51293
            - artifacts
              - git\_instructions.md
              - validation\_summary.md
              - worklog.log
            - ATXDocumentation
            - logs
            - subagents
            - mcp\_usage.json
            - metadata.json
            - logs
            - mcp-tools-cache.json
            - trust-settings.yaml
            - toapilambdanode16-comprehensive-cod...
            - todoapilambda-comprehensive-codebas...
            - scientific-processor
            - utilities

validation\_summary.md

```
1 # Validation Result Summary
2
3 ## TRANSFORMATION SUMMARY
4 BEFORE: The Spring-petclinic repository (Spring Boot 4.1.0, Java 17) contained no ATXDocumentation/
5 directory or comprehensive codebase analysis documentation.
6 AFTER: The transformation created a complete ATXDocumentation/ directory with 30 documentation files
7 (2,436 total lines) organized across 7 subdirectories (architecture/, analysis/, diagrams/,
8 migration/, reference/, specialized/, technical-debt/) plus 3 root-level files (README.md,
9 project-overview.md, technical-debt-report.md). The documentation covers system architecture, code
10 structure, all classes/interfaces/endpoints, dependency analysis, text-based diagrams, technical debt
11 with severity ratings, migration planning, and specialized documentation for database schemas, UI
12 components, and infrastructure. The technical-debt-report.md includes an AWS Transformation
13 Recommendation section recommending AWS/spring-boot-version-upgrade and AWS/java-version-upgrade. The
14 ATXDocumentation/ folder was copied to ~/.aws/atx/custom/20261005_151702_ffa51293/ATXDocumentation/
15 while preserving the original in the project root.
16
17 ## OVERALL STATUS: COMPLETE
18
19 ## EXIT CRITERIA RESULTS
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

o ( (.venv) ) participant: /workshop

Ln 1, Col 1 Spaces: 4 UTF-8 LF ( ) Markdown Layout: U.S.

Validation Summary and Transformation Details

**Validation Result Summary**

**## TRANSFORMATION SUMMARY**

BEFORE: The spring-petclinic repository (Spring Boot 4.1.0, Java 17) contained no ATXDocumentation/ directory or comprehensive codebase analysis documentation.

AFTER: The transformation created a complete ATXDocumentation/ directory with 30 documentation files (2,436 total lines) organized across 7 subdirectories (architecture/, analysis/, diagrams/, migration/, reference/, specialized/, technical-debt/) plus 3 root-level files (README.md, project-overview.md, technical-debt-report.md). The documentation covers system architecture, code structure, all classes/interfaces/endpoints, dependency analysis, text-based diagrams, technical debt with severity ratings, migration planning, and specialized documentation for database schemas, UI

Terminal Output:

```
((.venv) ) participant: /workshop$ AWS_DEFAULT_REGION=us-east-1 python3 utilities/invoke-api.py \
--endpoint "SAPI_ENDPOINT" \
--path "/jobs/batch" \
--data '{
  "batchName": "fleet-upgrade-wave-1",
  "jobs": [
    {
      "source": "https://github.com/spring-projects/spring-petclinic",
      "command": "atx custom def exec -n AWS/
java-version-upgrade -p /source/spring-petclinic --configuration \"additionalPlanContext=The target Java versio
n to upgrade to is Java 21.\" -x -t\"},
    {
      "source": "https://github.com/venuvasu/todoapilambda",
      "command": "atx custom def exec -n AWS/python-ver
sion-upgrade -p /source/todoapilambda --configuration \"additionalPlanContext=The target Python version to upgr
ade to is Python 3.13.\" -x -t\"},
    {
      "source": "https://github.com/venuvasu/toapilambdanode16",
      "command": "atx custom def exec -n AWS/nodejs
-version-upgrade -p /source/toapilambdanode16 --configuration \"additionalPlanContext=The target Node.js versio
n to upgrade to is 22.\" -x -t\"}
  ]
}'
{
  "batchId": "fleet-upgrade-wave-1-20261005-160109",
  "status": "PROCESSING",
  "totalJobs": 3
}
```

Amazon S3 Buckets

General purpose buckets (3)

| Name                                   | AWS Region                      | Creation date                         |
|----------------------------------------|---------------------------------|---------------------------------------|
| atx-access-logs-509399592491-us-east-1 | US East (N. Virginia) us-east-1 | October 5, 2026, 15:41:16 (UTC+05:30) |
| atx-output-509399592491-us-east-1      | US East (N. Virginia) us-east-1 | October 5, 2026, 15:41:33 (UTC+05:30) |
| atx-source-509399592491-us-east-1      | US East (N. Virginia) us-east-1 | October 5, 2026, 15:41:33 (UTC+05:30) |

**Account snapshot** Updated daily View dashboard

**External access summary** Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions

Account ID: 5093-9959-2491

Search

[Option+S]

United States (N. Virginia)

Lambda > Functions

1

Lambda

Dashboard

Applications

Functions

▼ Function resources

Capacity providers

Code signing configurations

Event source mappings

Layers

Replicas

▼ MicroVMs

MicroVM Images

MicroVMs

Network connectors

Functions (13)

Last fetched 05/10/2026, 21:36:40 GMT+5:30

Actions

Create function

Search by attributes or search by keyword

|                          | Function name                                      | Description                                                                              | Package type | Runtime     | Type     | Last modified |
|--------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------|--------------|-------------|----------|---------------|
| <input type="checkbox"/> | atx-upload-unt                                     | -                                                                                        | zip          | Python 3.12 | Standard | ago           |
| <input type="checkbox"/> | code-editor-SwapInstanceProfileLambda-UFlp6fOYYjZe | Point the Web IDE instance at the participant instance profile after bootstrap completes | Zip          | Python 3.13 | Standard | 6 hours ago   |
| <input type="checkbox"/> | code-editor-CheckSSMDocLambda-0Q7eKsZ7x2x0         | Check SSM document on EC2 instance                                                       | Zip          | Python 3.13 | Standard | 6 hours ago   |
| <input type="checkbox"/> | atx-terminate-job                                  | -                                                                                        | Zip          | Python 3.12 | Standard | 6 hours ago   |
| <input type="checkbox"/> | atx-submit-job                                     | -                                                                                        | Zip          | Python 3.12 | Standard | 6 hours ago   |
| <input type="checkbox"/> | code-editor-RunSSMDocLambda-ywuj7jrLH2Ft           | Run SSM document on EC2 instance                                                         | Zip          | Python 3.13 | Standard | 6 hours ago   |
| <input type="checkbox"/> | code-editor-SecretPlaintextLambda                  | Return the value of the                                                                  | Zip          | Python 3.13 | Standard | 6 hours ago   |

CloudShell

Agent Toolkit for AWS

Feedback

© 2026, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences

us-east-1.console.aws.amazon.com/batch/home?region=us-east-1#

Account ID: 5093-9959-2491

Search

[Option+S]

United States (N. Virginia)

AWS Batch > Dashboard

1

AWS Batch

Dashboard

Jobs

Job definitions

Job queues

Environments

Consumable resources

Scheduling policies

Wizard

▼ Console settings

Permissions

Jobs

Job definition

Job queues

Compute environment

Scheduling policies

Reset settings

Dashboard

Batch resources

Batch metrics - new

Auto-refresh

Dashboard last updated: 09:39:12 PM. Auto-refreshes every 60 seconds.

Jobs overview

| Runnable | Running | Succeeded | Failed |
|----------|---------|-----------|--------|
| 0        | 3       | 6         | 0      |

CloudShell

Agent Toolkit for AWS

Feedback

© 2026, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences



us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log...

CloudWatch > Log management > /aws/batch/atx-transform > atx/default/445b4d4891ab4c4893fe13a8d8a6b2f0

### CloudWatch

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

1m 1h UTC timezone

Display

| Timestamp                                                   | Message                                                                     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------|
| There are older events to load. <a href="#">Load more</a> . |                                                                             |
| 2026-10-05T16:07:36.789Z                                    | + 14:                                                                       |
| 2026-10-05T16:07:36.790Z                                    | ● Completed in 1ms                                                          |
| 2026-10-05T16:07:43.304Z                                    | \ Using tool: grep (trusted)                                                |
| 2026-10-05T16:07:43.304Z                                    | :                                                                           |
| 2026-10-05T16:07:43.304Z                                    | ● Pattern: java.version.*17 java.*17 jdk.*17 JDK.*17                        |
| 2026-10-05T16:07:43.304Z                                    | ● Path: /source/spring-petclinic                                            |
| 2026-10-05T16:07:43.304Z                                    | ● Include: *.xml,*.yaml,*.yml,*.gradle,*.properties,*.json,*.md,Dockerfile* |
| 2026-10-05T16:07:43.307Z                                    | :                                                                           |
| 2026-10-05T16:07:43.307Z                                    | ● Completed in 3ms                                                          |
| 2026-10-05T16:07:51.292Z                                    | \ Using tool: shell (trusted)                                               |

CloudShell Agent Toolkit for AWS Feedback © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

d3dw1a2c3vaewn.cloudfront.net/?folder=%2Fworkshop

validation\_summary.md

```
1 # Validation Result Summary
2
3 ## TRANSFORMATION SUMMARY
4 **BEFORE:** A Python 3.8 serverless Todo API Lambda application using `typing.List`, `typing.Dict`,
5 `typing.Optional`, `typing.Union` for type hints, `datetime.utcnow()` for timestamps, and `Runtime:
6 python3.8` in the SAM template. Documentation referenced Python 3.8 throughout.
7
8 **AFTER:** The application was upgraded to Python 3.13. Key changes across 13 files:
9 - SAM template `Runtime` updated from `python3.8` to `python3.13` (template.yaml)
10 - Type hints modernized: `List` -> `list`, `Dict` -> `dict`, `Tuple` -> `tuple`, `Optional[X]` ->
11 `X | None`, `Union[X, Y]` -> `X | Y` across all source files (advanced_processor.py, analytics.py,
12 batch_processor.py, lambda_function.py, mock_todo_service.py, models.py, protocols.py, todo_service.
13 py, validation.py)
14 - `Iterator` import moved from `typing` to `collections.abc` (batch_processor.py)
15 - Deprecated `datetime.utcnow()` replaced with `datetime.now(timezone.utc)` (analytics.py,
16 batch_processor.py, lambda_function.py, models.py)
17 - Unused typing imports (`List`, `Dict`, `Optional`, `Union`, `Tuple`) removed
18 - README.md updated to reference Python 3.13
19 - requirements.txt comments updated to document Python 3.13 features
20 - All 32 unit tests pass on Python 3.13.13
21
22 ## OVERALL STATUS: COMPLETE
23
24 ## EXIT CRITERIA RESULTS
25
26 **Criterion:** 1. All code runs successfully on the target Python version (3.11, 3.12, or 3.13 as
27 specified) without syntax errors.
28
29 **Verification Method:** Compiled all 10 Python source files with Python 3.13.13 using `py_compile`,
30 and ran the full test suite with `/tmp/venv313/bin/pytest`.
31
32 **Status:** PASS
33
34 **Evidence:** All source files compiled without errors under Python 3.13.13: `ALL COMPILED OK`. The
35 test suite ran on `platform linux -- Python 3.13.13, pytest-9.1.1` and all 32 tests passed: `32
36 passed in 0.22s`. No syntax errors were reported.
37
38 **Observations:** None
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Markdown Layout: U.S.

